

Splatt3R-SLAM: From Image Pairs to Persistent Maps

Fast local predictions.
One map that survives pose correction.

Zelong Li
MSc Computer Science | University of Amsterdam

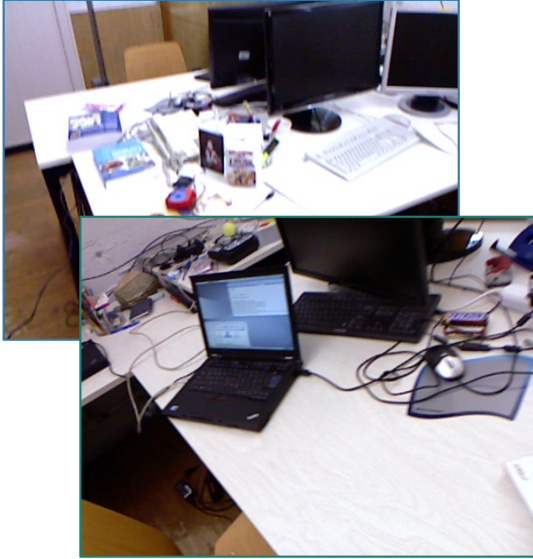
Supervisors: Prof. Martin Oswald | Dr. Qi Zhang



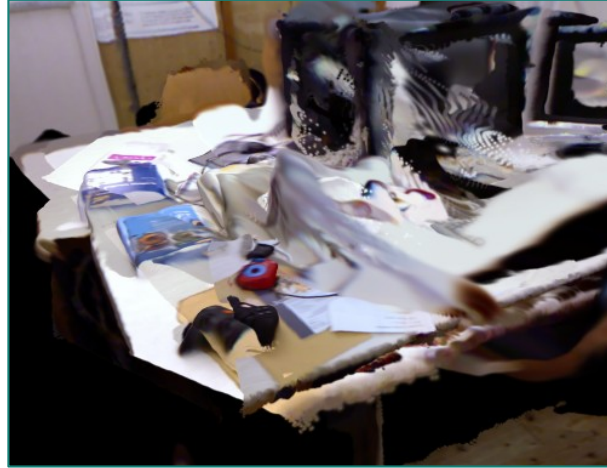
A pair is not a map

The gap between a local prediction and a lasting scene

ONE IMAGE PAIR



ONE LOCAL MAP



Useful immediately

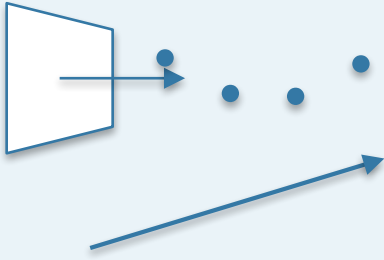
A VIDEO SEQUENCE



Many local frames
Many overlapping surfaces

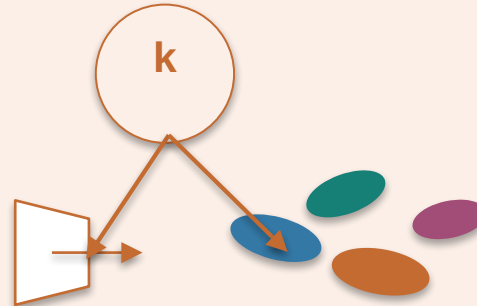
The research question: how can pairwise predictions become one persistent map?

REVISIT



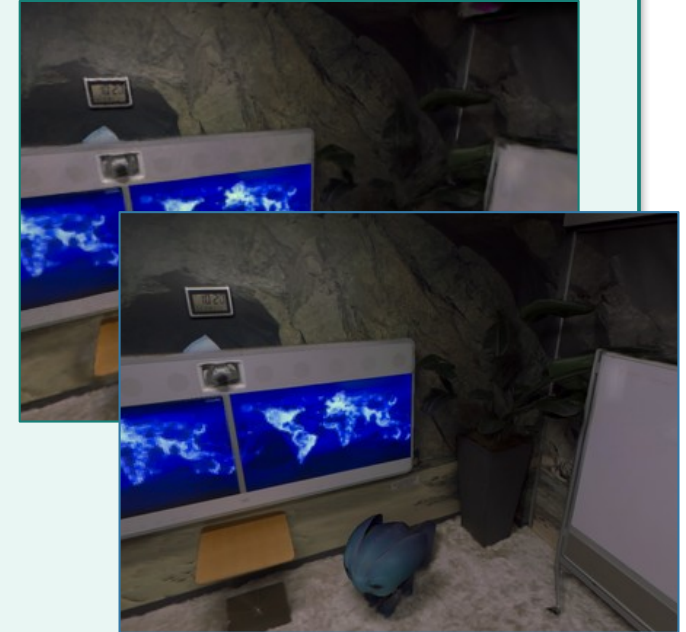
A later camera must still localise.

CORRECT



Loop closure must move old content.

RENDER



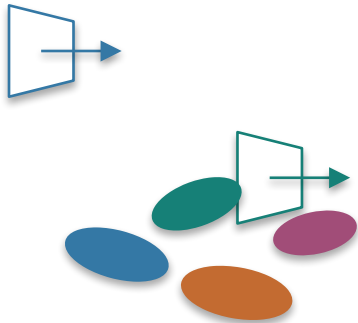
A new view must look like the scene.

Tracking, correction, and rendering must agree on the same scene.

Why direct accumulation fails

The sequence adds problems that a two-view model never sees

1 Different coordinates



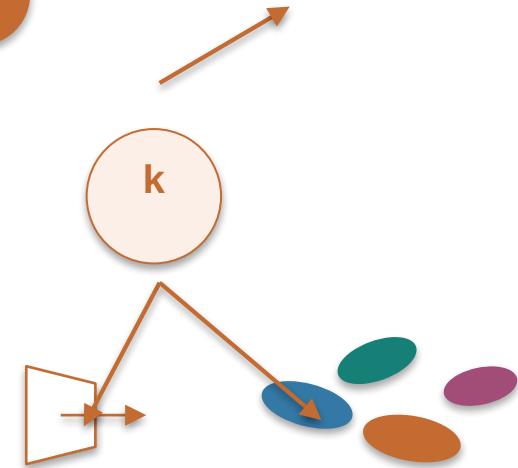
Each pair predicts in its own camera frame.

2 Repeated surfaces



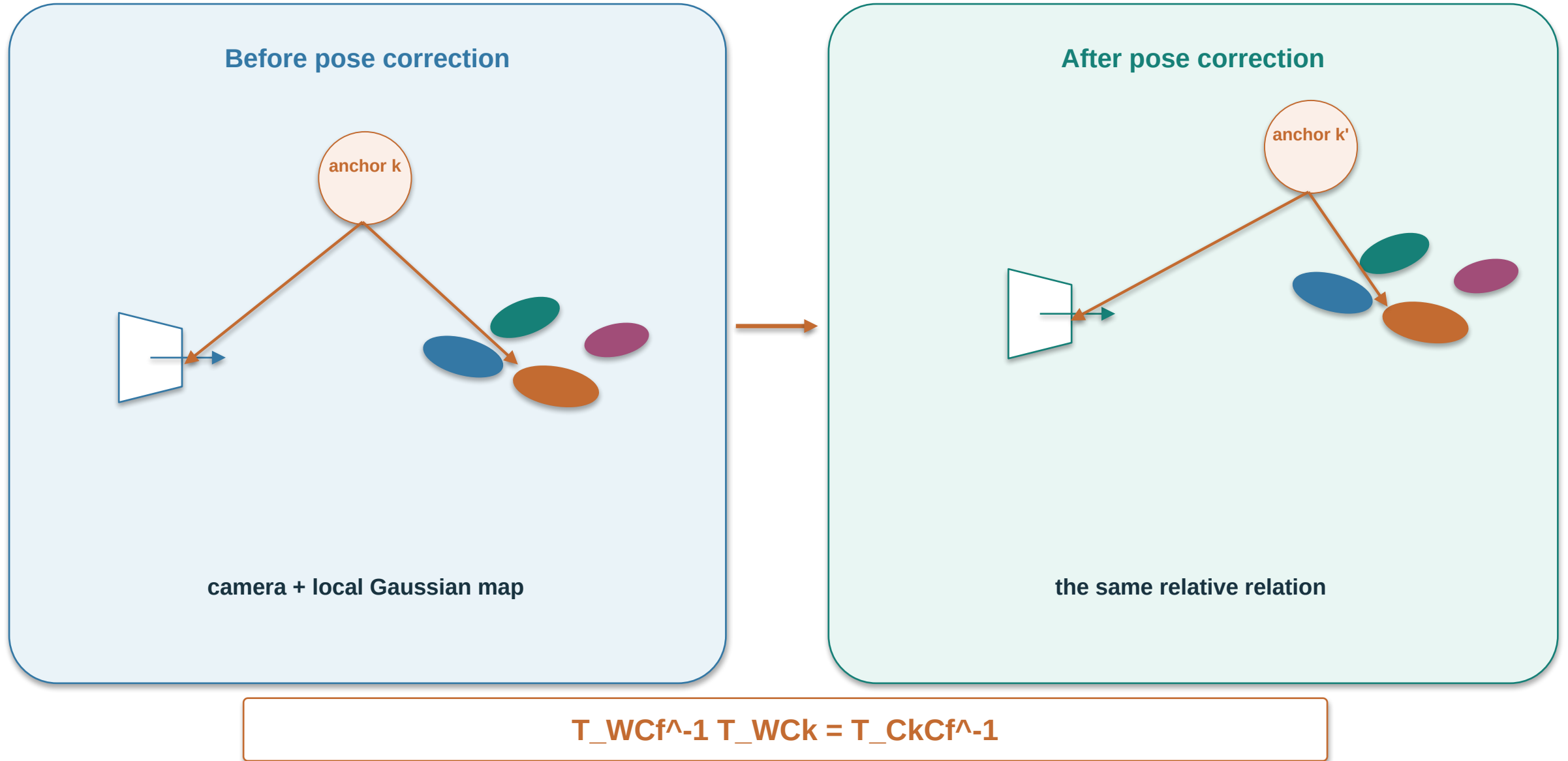
Several keyframes place colour and opacity on one ray.

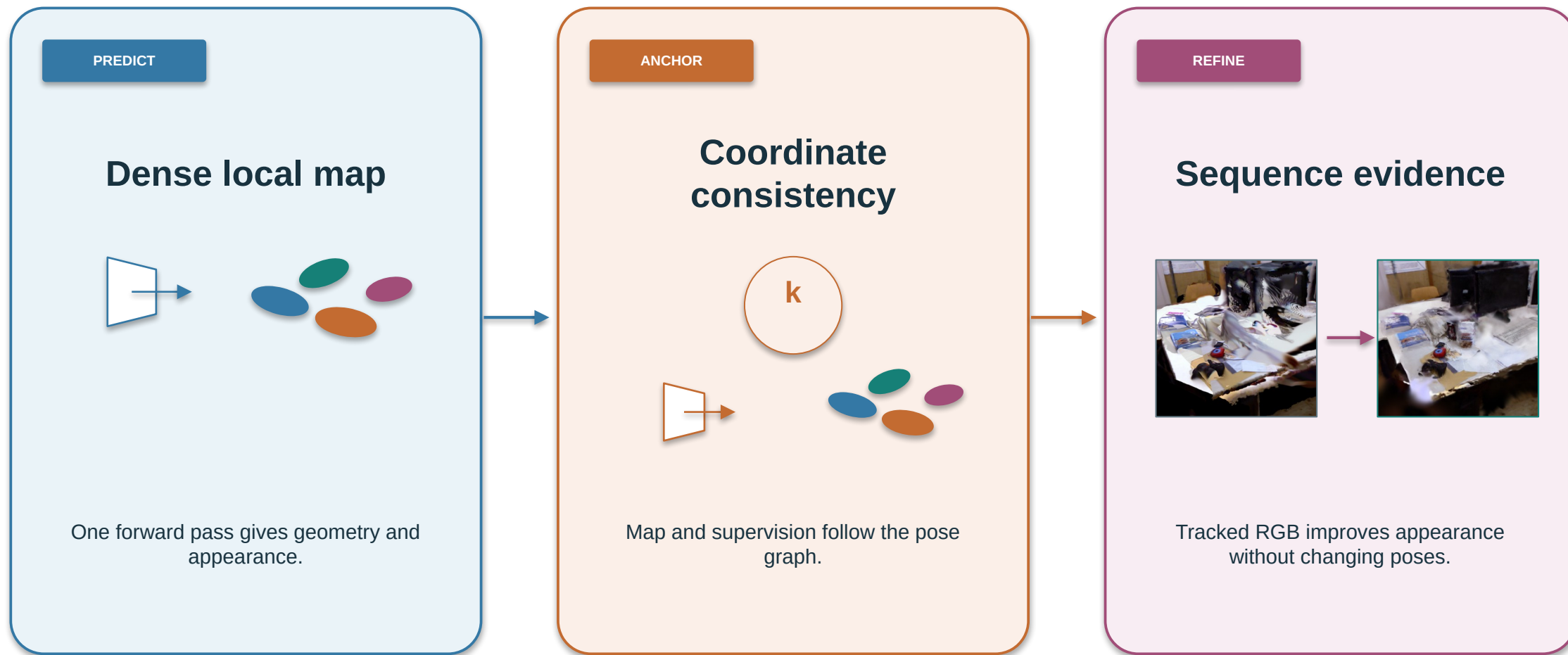
3 Pose corrections



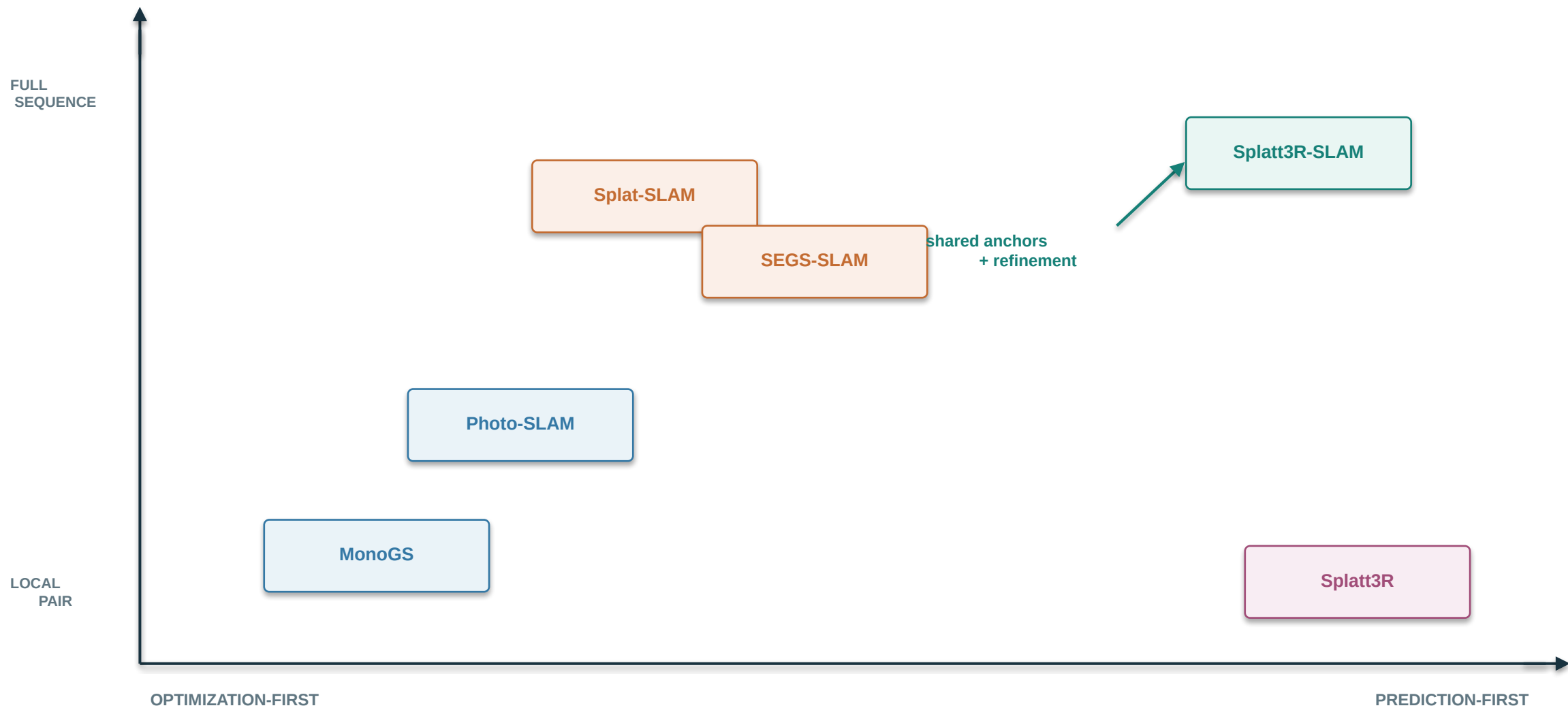
Loop closure changes where old predictions belong.

Concatenation is not sequence-level mapping.





The experiments ask whether each step preserves the claim.



Positioning diagram only. Quantitative ranking requires a common protocol.

(a) Predict

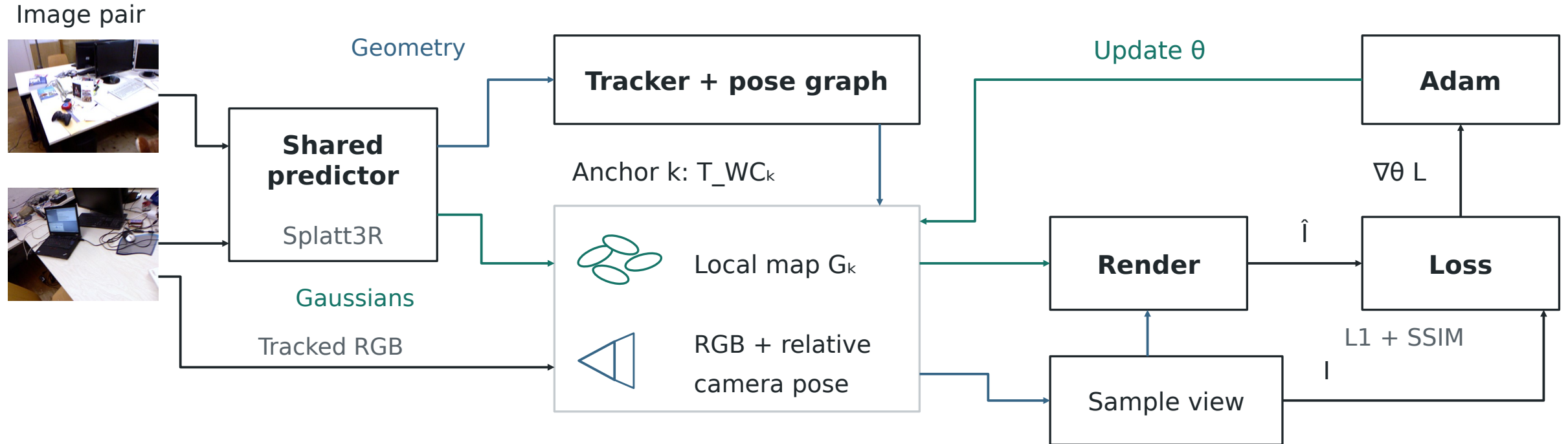
Ch. 2 · Eq. (2.7)

(b) Anchor

Ch. 4 · Eqs. (4.1–4.3)

(c) Refine

Ch. 7 · Eq. (7.1)



Before refinement

After refinement

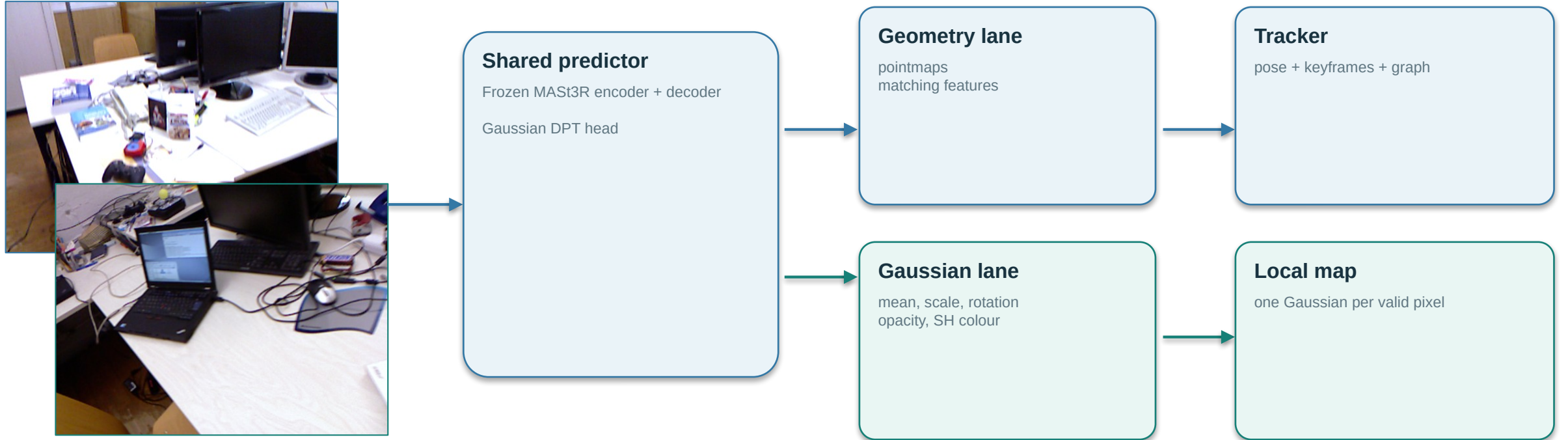


Shared-anchor relation

$$T_{wcf}^{-1} T_{wck} = T_{ckcf}^{-1}$$

Pose corrections move map and cameras together.

Optional initialisation: Chs. 5–6 · Eq. (6.1)



$$\mu_n = x_n + \Delta_n \quad | \quad \Sigma_n = R(q_n) \text{diag}(s_n^2) R(q_n)^T$$

Paper Sec. III-A / Eq. (1) | Thesis Ch. 2 / Eq. (2.7)

Why the supervision camera shares the anchor

Coordinate control and refinement loop

Before a loop closure

Local map and camera use keyframe k.
Their relative transform is fixed.



$$T_{WCf}^{-1} T_{Wck} = T_{CkCf}^{-1}$$

Pose correction

The pose graph changes T_{Wck} .
The map and camera move together.



shared anchor: relation preserved

Controlled 6% scale test

Fixed world camera: refinement reverses the correction.
Shared anchor: correction remains after about 500 steps.

Desk + room controls

REFINEMENT LOOP

sample RGB

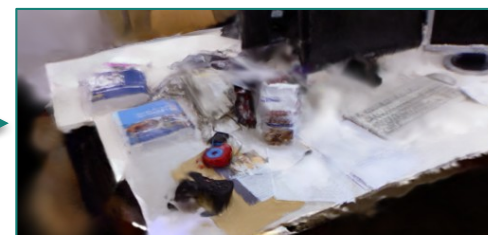
compose pose

render map

L1 + SSIM

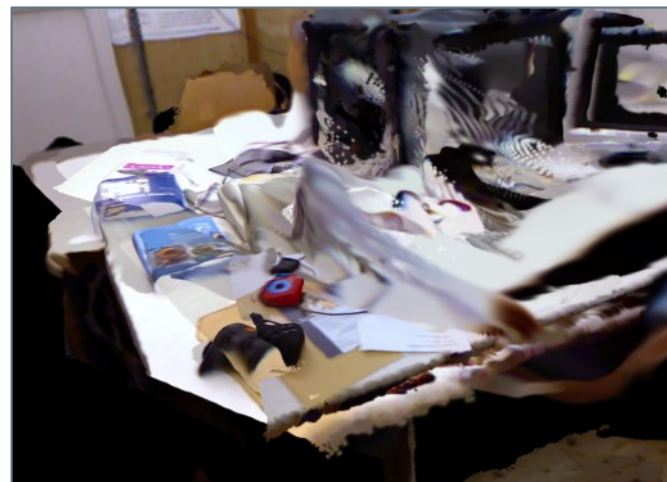
Adam update

Only Gaussian scale, rotation, opacity, and DC color receive gradients.



Refine: use the sequence without moving the tracker

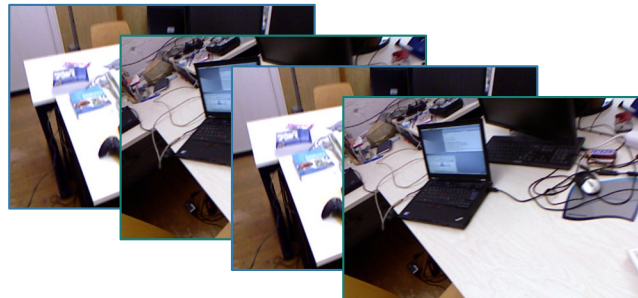
The real control loop is sampling, rendering, and Adam



same camera



SUPERVISION POOL



Reservoir + recent ring

200 historical | 64 recent
30% recent sampling

Render

compose live anchors
render one RGB view

Loss + Adam

0.8 L1 + 0.2 D-SSIM
update appearance

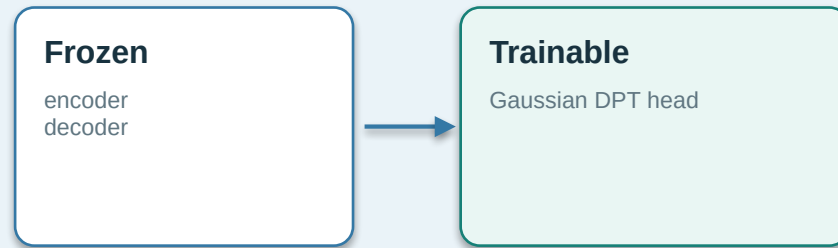
repeat with another tracked view

Gradients update Gaussian appearance. They never update poses or tracking weights.

Paper Sec. III-C / Eq. (5) | Thesis Ch. 7 / Eq. (7.1)

PREDICTION

Head-only adaptation

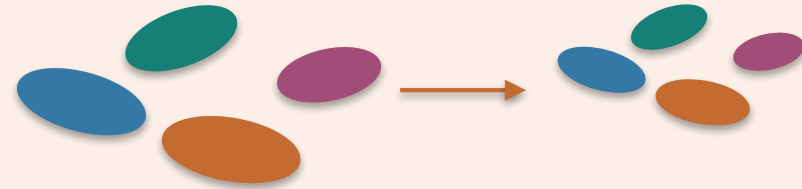


Changes pairwise appearance
without changing tracking features

MSE + 0.25 LPIPS-VGG

INSERTION

Opacity attenuation



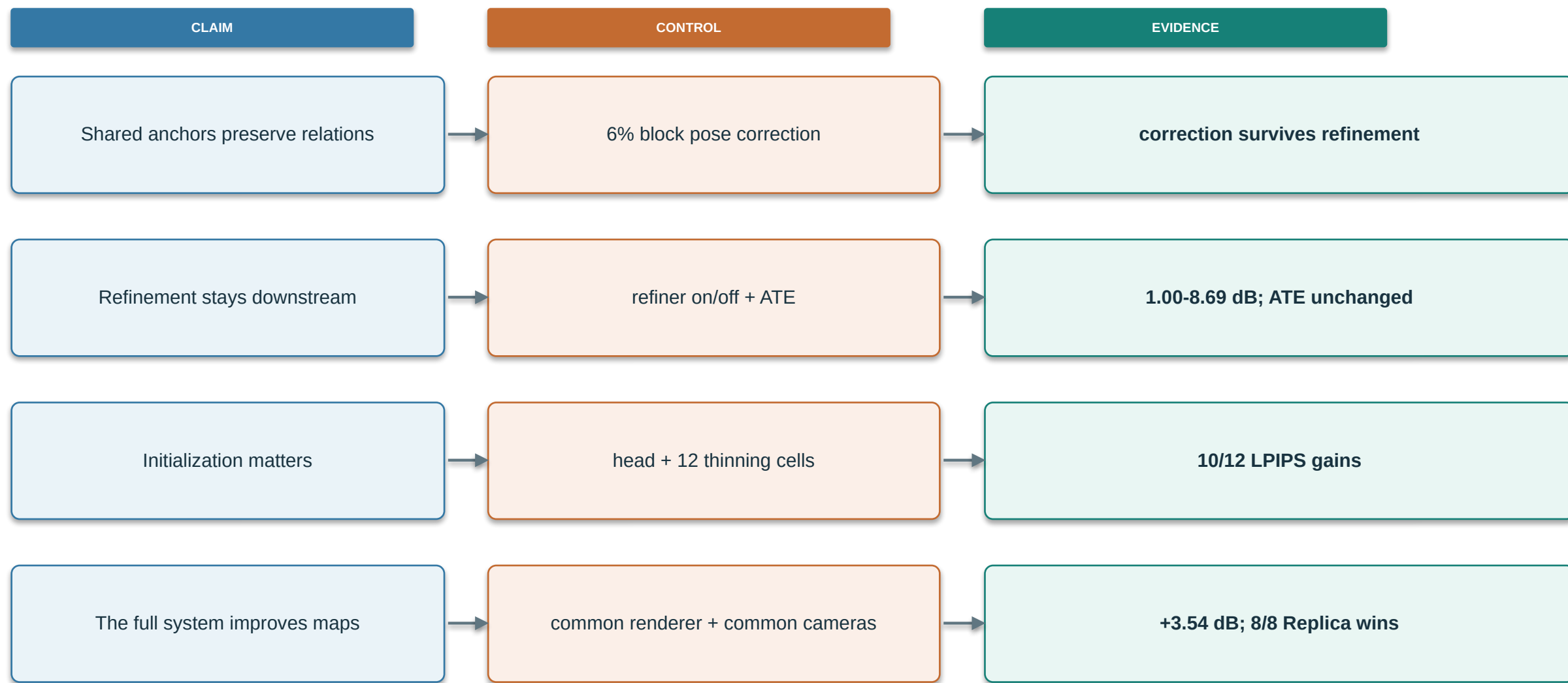
Lower initial opacity
for low-confidence predictions

$\alpha' = \alpha \text{ clip}(1 - 2 \lambda (1 - \text{rank}), 0.1, 1)$

External baseline comparison: released head, attenuation off.

Every claim has a matching experiment

The evaluation follows the theory, not a list of unrelated scores



RQ1-RQ4 in the thesis use this same chain.

A common renderer makes the comparison meaningful

Same target cameras, resolution, alignment, and metrics

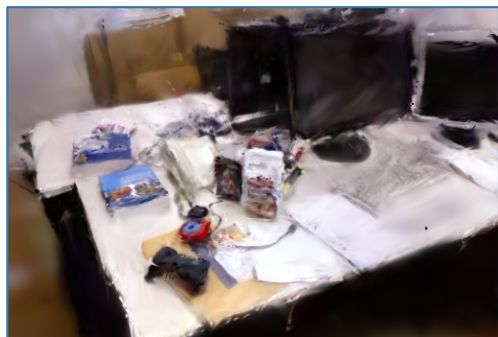
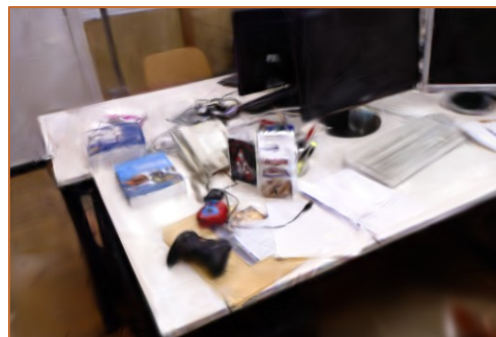
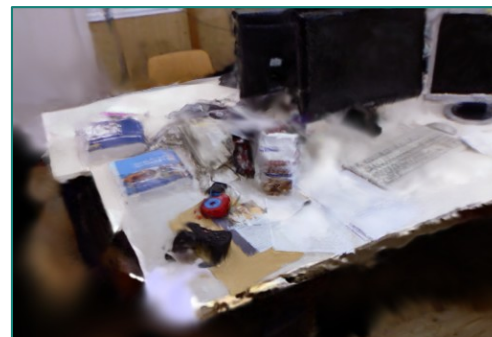


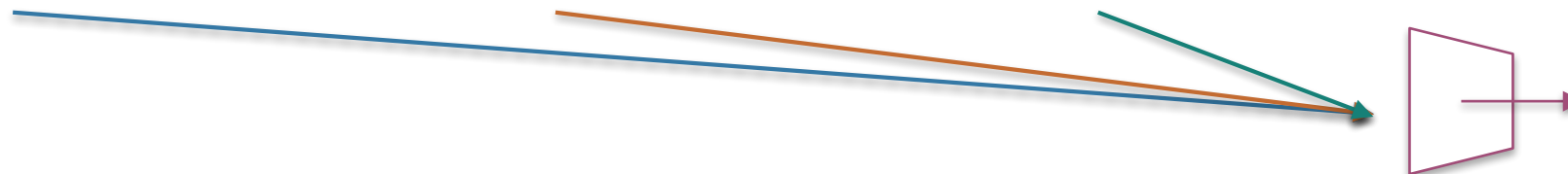
Photo-SLAM



MonoGS



Splatt3R-SLAM



one GT camera

100 frames

outside the union of keyframes

Full SH

all exported colour bands

Same image size

512x288 or 512x384

Same metrics

PSNR + LPIPS-AlexNet

Replica: 8 scenes | TUM desk: 3 renderable systems | TUM: 9-sequence ATE

23.02 dB

Splatt3R-SLAM mean

+3.54 dB

over Photo-SLAM

8 / 8

scene PSNR wins

5 / 8

scene LPIPS wins

MEAN PSNR (dB)

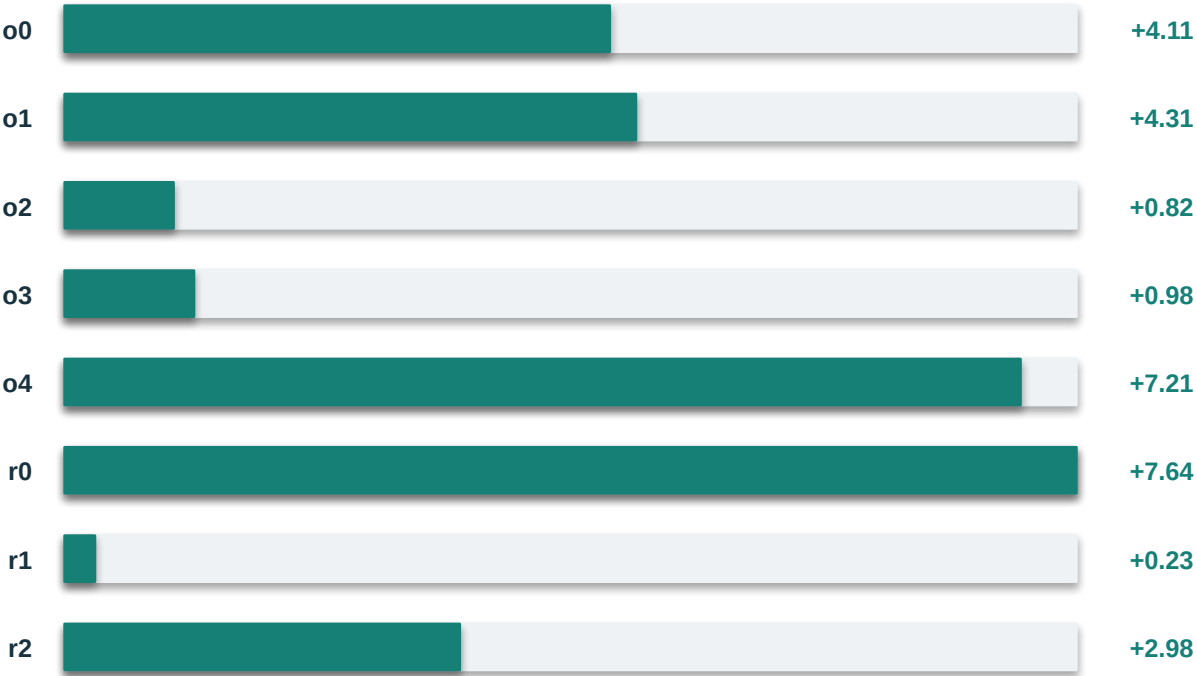


Photo-SLAM



Splatt3R-SLAM

OURS - PHOTO-SLAM, PER SCENE



Smallest margin: room1 +0.23 dB | largest: room0 +7.64 dB

Replica reconstruction across scenes

Four representative scenes | common cameras



All images use the same selected camera within each scene.

Qualitative reconstruction at common cameras

Measured renders; identical crops and scales

Ground truth

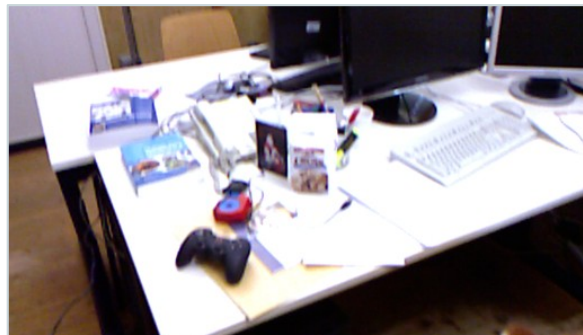
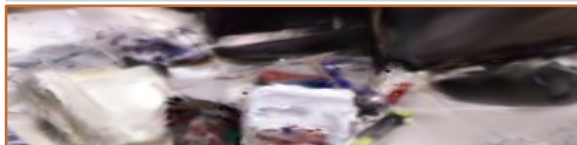
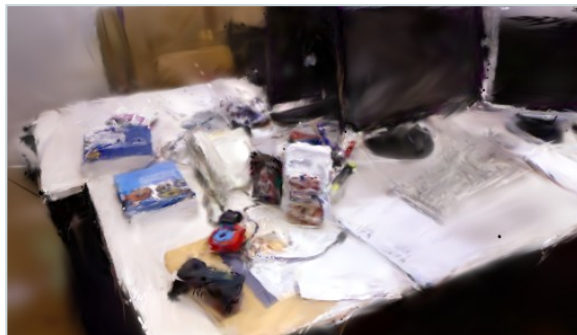
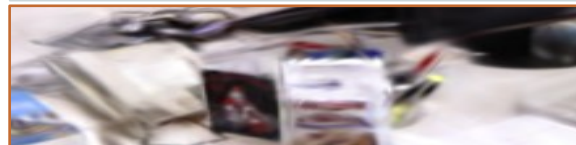
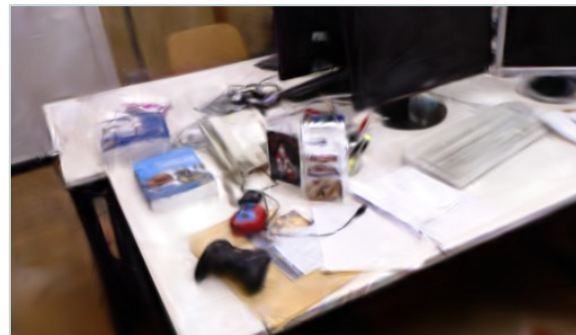


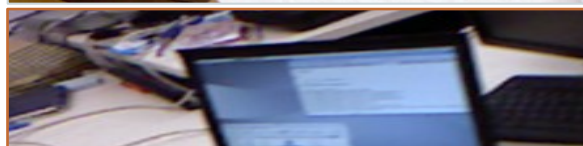
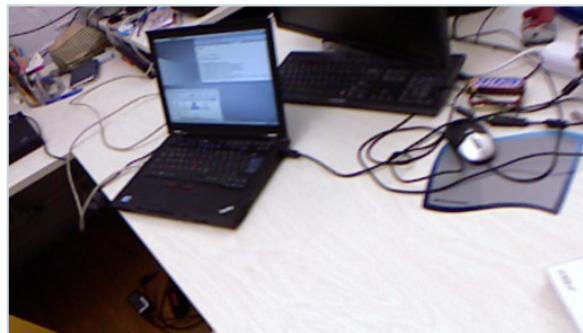
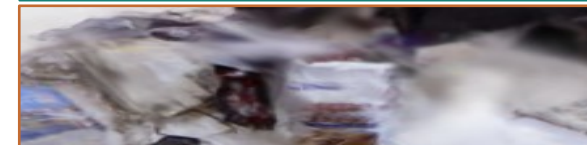
Photo-SLAM



MonoGS



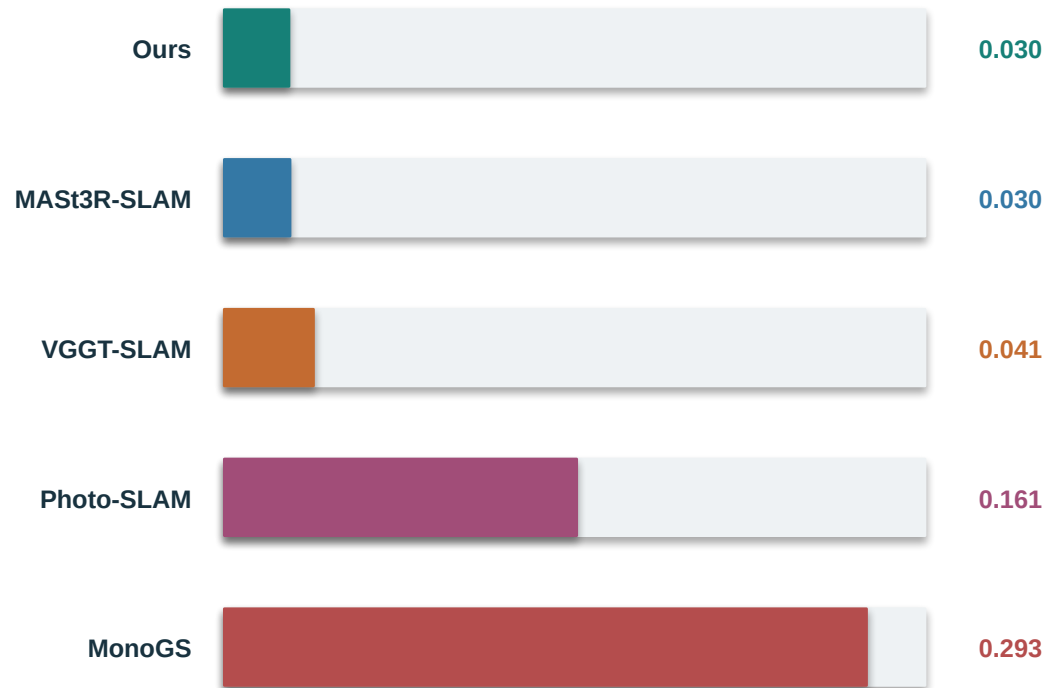
Ours



TUM desk: two middle-ranked Ours-minus-MonoGS PSNR views

Orange panels are identical pixel crops across all methods.

MEAN ATE RMSE (m) - LOWER IS BETTER



What this result proves

- The mapping path does not disturb tracking.
- Ours and MAST3R-SLAM have the same rounded mean.

What it does not prove

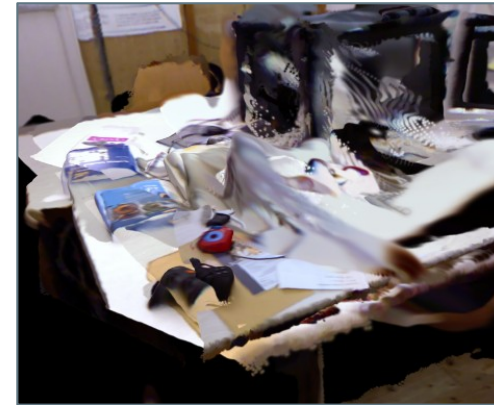
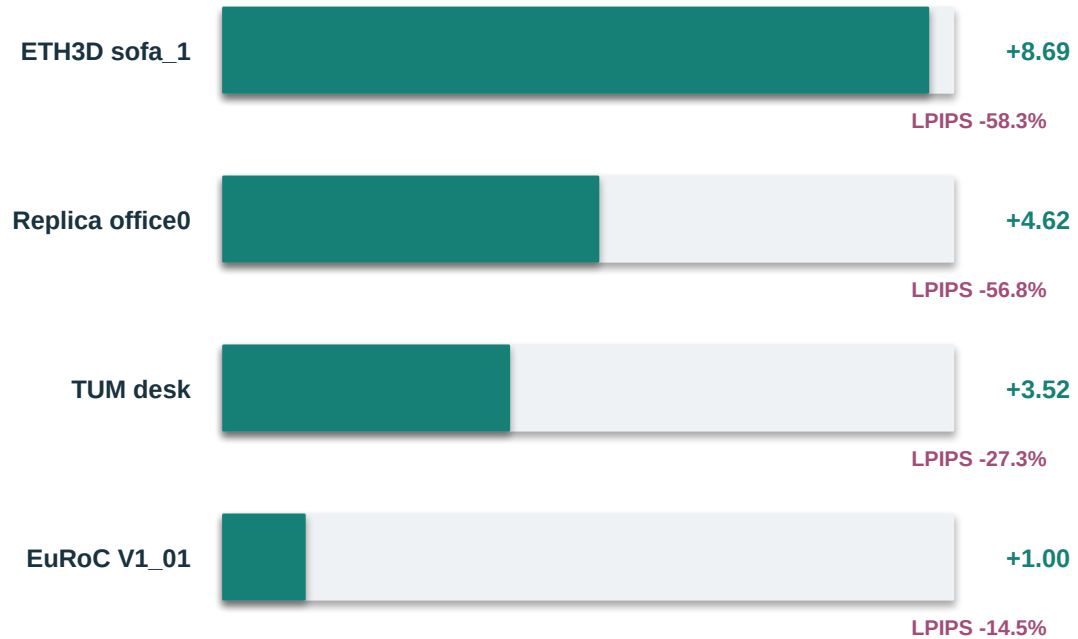
- Tracking is inherited, not a new contribution.
- Single runs can hide system variance.

The map improves appearance downstream of pose estimation.

Refinement gives the largest controlled gain

Same head, same insertion, refiner off versus on

PSNR GAIN (dB)



refiner off



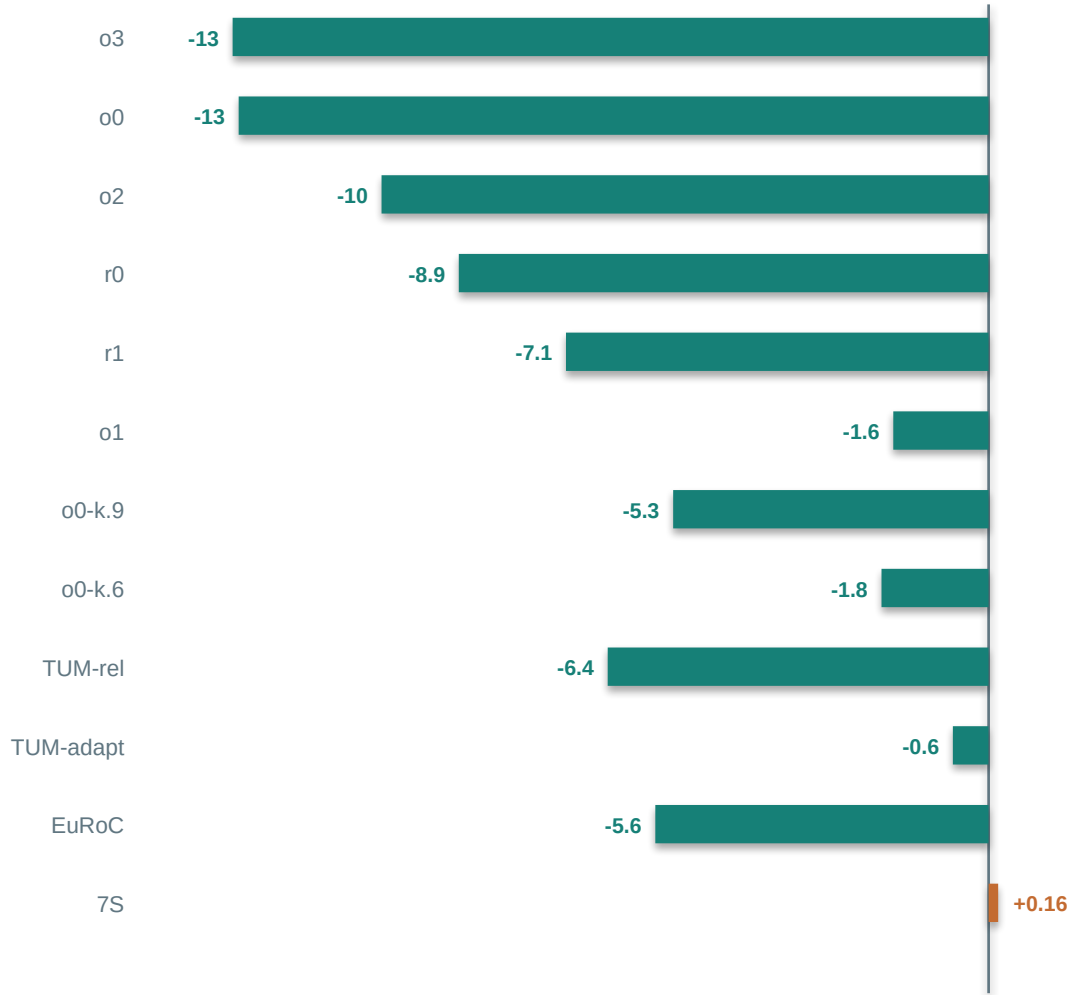
refiner on

Why the control is causal

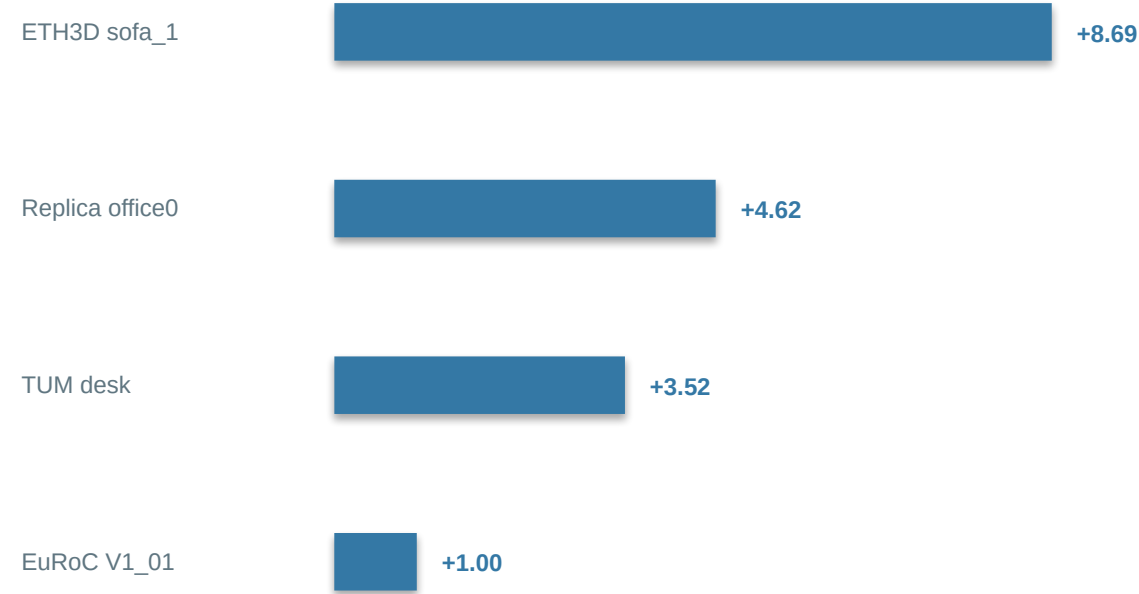
Only the refiner changes.
The tracker and Gaussian head stay fixed.

Result: sequence observations improve a feed-forward map.

Opacity attenuation: relative LPIPS change (%)



Refiner on vs. off: PSNR gain (dB)



Interpretation

Refinement gives the largest controlled gain.
Opacity reduction is robust; confidence allocation is not.

External baseline comparison uses the released head with attenuation off.



Second GPU

103 ms
tracker p50

+2.15 dB
causal gain

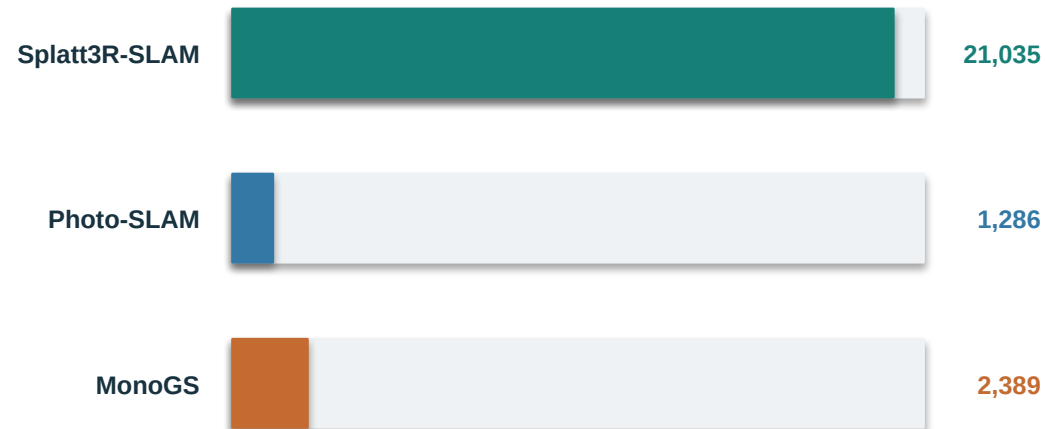
Shared GPU

206 ms
tracker p50

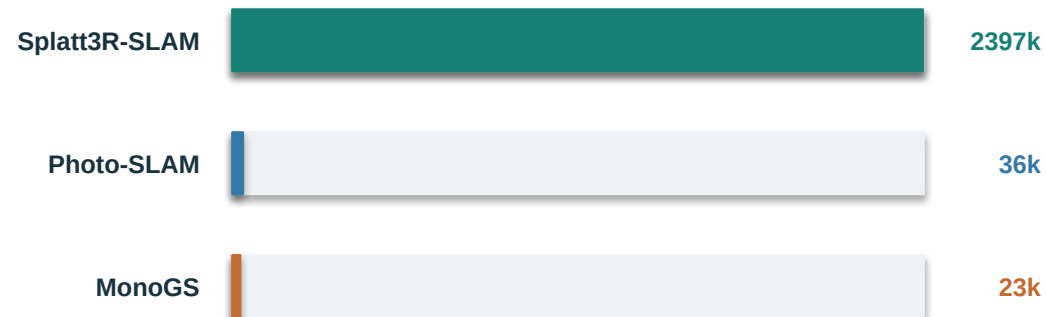
4.5 fps
sustained

Most of the desk gain appears by about 500 steps.

PEAK GPU MEMORY (MiB)



EXPORTED PRIMITIVES



Quality versus compactness

13.95

PSNR | full map

2.397M

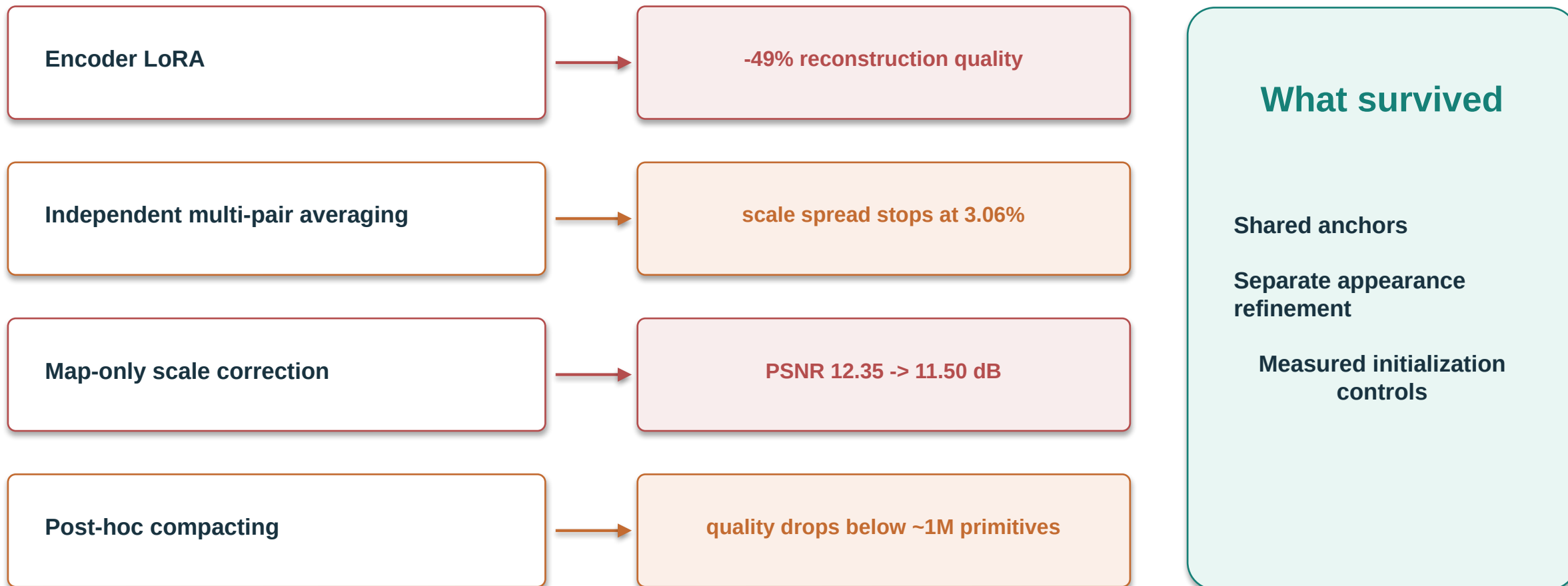
Gaussians

Post-hoc truncation collapses quality
below about one million primitives.

The next engineering target is not more
Gaussians.

It is quality per primitive.

The representation is effective, but not yet compact.



The useful theory is coordinate consistency, not a universal geometry correction.

Measured result

23.02

ours | dB

19.48

Photo-SLAM | dB

+3.54

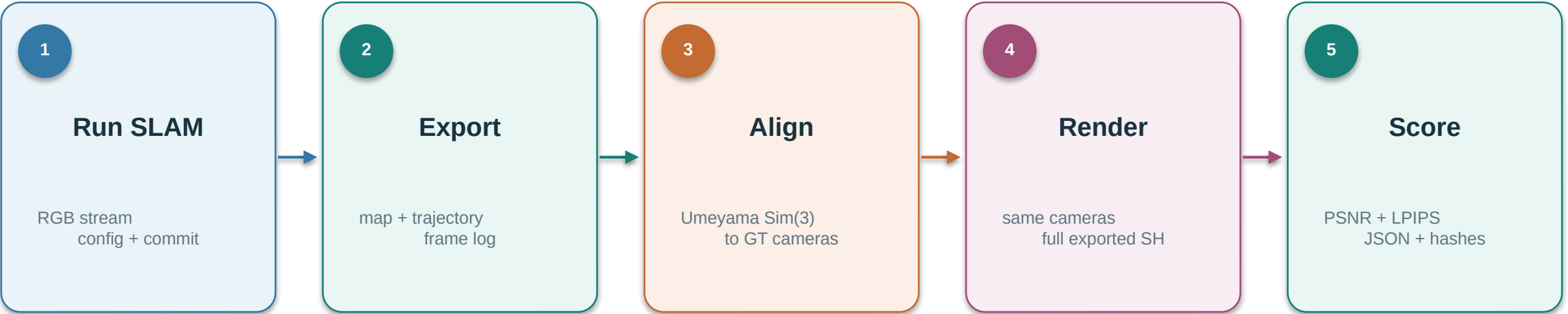
mean gain | dB

- Higher PSNR on every Replica scene.
- Better mean LPIPS: 0.135 versus 0.163.
- All values come from saved maps and one evaluator.

Common protocol

- **Input** RGB-only monocular
- **Coverage** all 8 Replica scenes
- **Views** 100 common non-keyframes per scene
- **Scoring** same cameras, renderer, resolution, metrics

Splatt3R-SLAM achieves state-of-the-art performance



`logs/paper_ral_20261007/<scene>/metrics.json`
stores per-frame MSE, LPIPS, camera alignment, selected views, and SHA256 hashes

RA-L PDF
7 pages

Review PDF
anonymous

MSc thesis
65 pages

Source ZIP
clean rebuild

Build: `cd docs/Thesis && ./build.sh [ral|review|master]`

One idea connects the system

Give the local map and its observation cameras the same keyframe anchor.

PREDICT

Dense local Gaussians in one pass

ANCHOR

Pose corrections move map and cameras together

REFINE

Sequence RGB improves appearance downstream



+3.54 dB

Replica

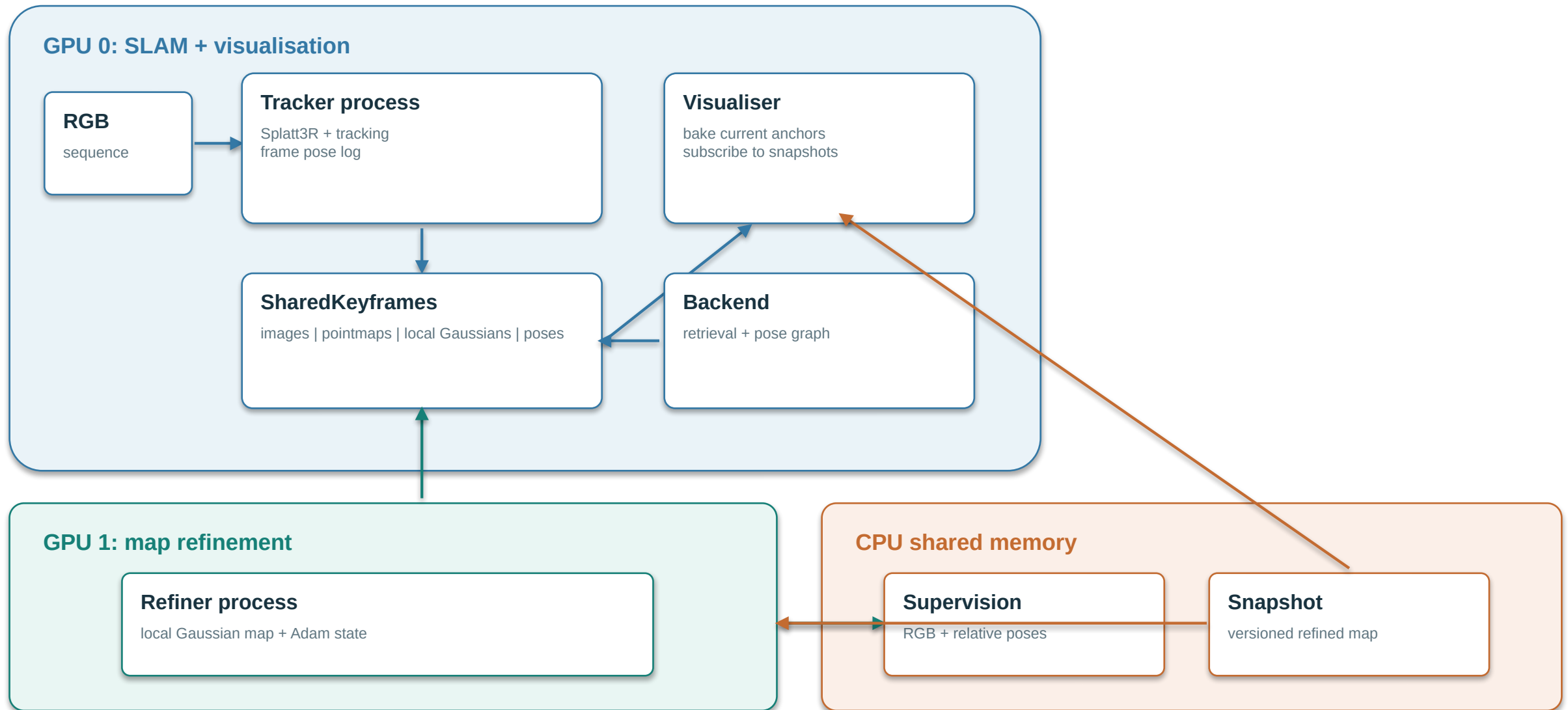
8 / 8

PSNR wins

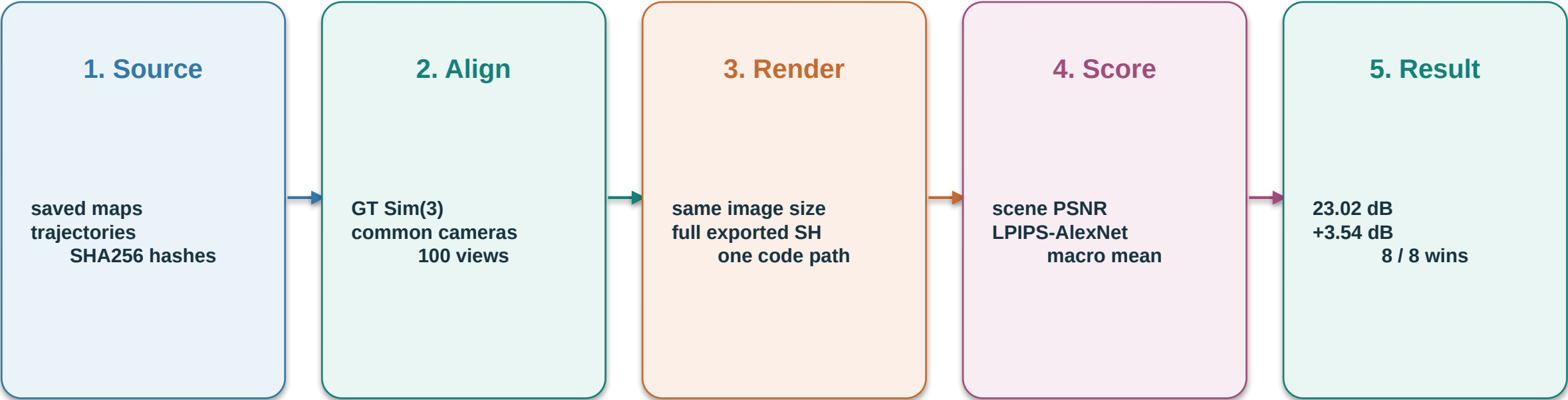
23.02

mean PSNR

Next: retain this quality with a compact causal map.



Tracking owns poses. Refinement publishes maps. No cross-process partial state.



Claim: Splatt3R-SLAM achieves state-of-the-art performance.

8
Replica scenes

800
scored views

8 / 8
PSNR wins

Backup: settings that define the result

The headline number is one explicit configuration

FIELD	VALUE	WHY IT MATTERS
Input	monocular RGB	all systems in the primary table
Head	released Splatt3R	no family adaptation
Insertion	attenuation off	lambda = 0
Centres	fixed	appearance-only external polish
Polish	300 seconds	post-sequence
Evaluation	100 non-keyframes	common GT cameras
Colour	full exported SH	degree differs by system
Metrics	PSNR + LPIPS-AlexNet	scene macro mean

Changing any row defines a different experiment.